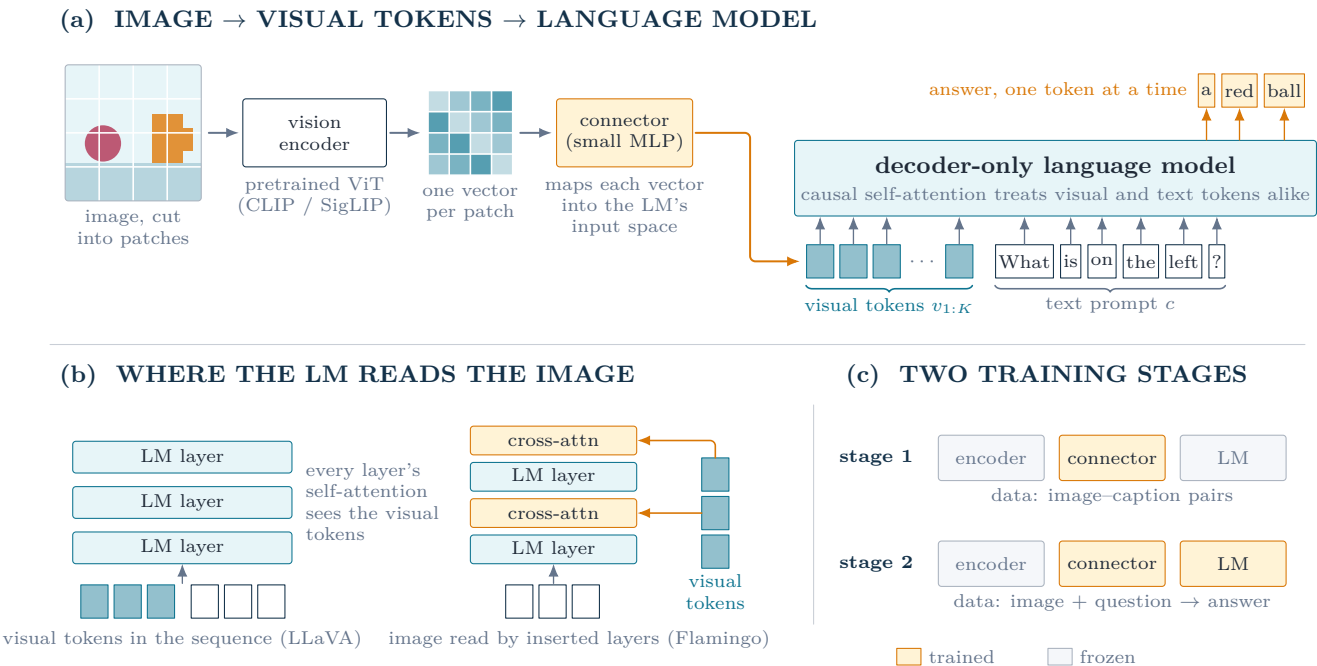
(a) IMAGE → VISUAL TOKENS → LANGUAGE MODEL
image, cut into patches
vision encoder
pretrained ViT (CLIP / SigLIP)
one vector per patch
connector (small MLP)
maps each vector into the LM's input space
answer, one token at a time
a
red
ball
decoder-only language model
causal self-attention treats visual and text tokens alike
What
is
on
the
left
?
visual tokens $v_{1:K}$
text prompt c
(b) WHERE THE LM READS THE IMAGE
LM layer
LM layer
LM layer
every layer's self-attention sees the visual tokens
visual tokens in the sequence (LLaVA)
cross-attn
LM layer
cross-attn
LM layer
visual tokens
image read by inserted layers (Flamingo)
(c) TWO TRAINING STAGES
stage 1
encoder
connector
LM
data: image–caption pairs
stage 2
encoder
connector
LM
data: image + question → answer
trained
frozen